

SUNRITA SARKAR

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EDUCATION

Vellore Institute of Technology, Chennai Expected Graduation, May 2027

Bachelor of Technology in Computer Science and Engineering 8.06

Concentrations: Object-Oriented Programming, Computing in Python, Java, C++, Full stack Web development

- **Related Coursework:** Statistics and Applications, Discrete Mathematics, Compiler Design, Computer Networks, Blockchain, Embedded System, Healthcare and Data Analytics, Cloud Computing, Fog and Edge Computing, AWS Fundamentals

Calcutta Girls' High School, Kolkata 2021-2023

12th Boards, Indian School Certificate 93.5%

- **Stream:** Physics, Chemistry, Mathematics with Computer Science

Calcutta Girls' High School, Kolkata 2009-2021

10th Boards, Indian Certificate of Secondary Education 96%

- **Stream:** Science with Computer Science

PROJECTS AND CERTIFICATIONS

- **Travel Website (Rael):** Developed a responsive travel web application to streamline trip planning and destination exploration, using HTML, CSS, and JavaScript. Designed an intuitive user interface with dynamic content rendering to enhance user experience and navigation. Implemented interactive features for browsing destinations and simulating booking workflows, focusing on usability and frontend performance. Structured the application with clean, maintainable code, demonstrating strong fundamentals in web development, UI design, and client-side logic.
- **Game AI for Tic-Tac-Toe (Minimax Algorithm):** Developed an interactive game application using Python, integrating an AI agent powered by the Minimax algorithm for optimal decision-making. Implemented modular game logic with efficient state management using NumPy and designed a responsive GUI using Pygame for real-time user interaction. Optimized recursive search performance using pruning techniques and structured the application for scalability, demonstrating strong problem-solving, algorithmic thinking, and software design principles relevant to full-stack and backend development.
- **Arrhythmia Detection using CNN on ECG Signals:** Developed a deep learning-based system to automatically detect cardiac arrhythmias from wearable ECG data using Convolutional Neural Networks (CNN). Performed signal preprocessing, feature extraction (RR intervals, morphological and spectral features), and segmented ECG signals for training. Utilized the MIT-BIH Arrhythmia Dataset and built a multi-class classification model to identify normal, atrial, and ventricular arrhythmias with high accuracy.

- **Blockchain-Based Agricultural Supply Chain Transparency System:** Designed and implemented a secure supply chain framework integrating blockchain, AI, and cryptography to ensure transparency and traceability of agricultural produce. Developed an anomaly detection module using Isolation Forest to identify irregular production and transport patterns, and applied SHA-256 hashing with a private blockchain to enable tamper-proof data storage and verification.
- **Automatic Coloured Bottle Detection & Rejection System (Arduino):** Developed an automated quality control system using Arduino and LDR sensors to detect coloured bottles based on light transmission principles. Designed a threshold-based detection algorithm to differentiate between transparent and defective bottles and trigger real-time alerts using a buzzer. Built a cost-effective, sensor-based solution suitable for industrial automation and small-scale production lines.
- Oracle Certified Professional: Java SE 17 Developer Certification
- International Astronomy and Astrophysics Competition 2025 Finalist

SKILLS

Programming: Java, C, C++, Python, HTML/CSS, JavaScript, React.js, Assembly Language Programming

Technical/Programmatic: Data Structure and Algorithms, Design and Analysis of Algorithms, Digital Painting

Databases: SQL(MySQL)

Softwares: Solidworks, Blender, Adobe Sketchbook, Ibis Paint

Platforms: Windows, Linux

Languages: English, Bengali, Hindi, Spanish

LEADERSHIP

- **Treasurer – Bengali Literary Society, VIT Chennai**
Managed budgeting and financial operations while organizing and actively participating in cultural events, ensuring smooth execution and coordination.
- **Official Ambassador – International Astronomy and Astrophysics Competition**
Promoted the competition, coordinated outreach, and engaged students to increase participation and awareness.
- **Organizer, Anchoring Committee – Vibrance'26 (VIT Chennai Cultural Fest)**
Hosted and organized large-scale fest events, managing scripts, stage flow, and live audience engagement for seamless execution.